

Branding & Advertising

MGT 100 Week 7

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What is a brand?



When I say brand, most people think about logos. What are the hidden meanings in these images?

Pre-2010



October 6, 2010



October 12, 2010



The Gap adopted its iconic logo in 1969. It launched a multimillion-dollar redesign in October 2010, then scrapped it six days later after a backlash. Why scrap it?

Pre-2012



Dec. 10, 2012



Dec. 15, 2012



"While I believe the design element in question would win wide acceptance over time, it also is important that we listen to and respect what has been a significant negative response by students, alumni and other members of our community."
--Steven Dooley, UC SVP-External Relations

UC adopted its seal in 1884. We paid a relatively paltry \$386k for this new logo, but we only kept it for five days. Why? The current wordmark was rolled out more quietly in 2013

NBC Bay Area (2012) | 99% Invisible (2012) | UCOP Wordmark



Westinghouse logo, licensed by CBS to

- Westinghouse Electric Co. (nuclear energy)
 - Westinghouse Digital LLC (LCD and LED televisions)
 - TECO-Westinghouse Motor Group (electric motors)
 - Westinghouse Lighting Corporation (light bulbs and fixtures)
 - Westinghouse Solar
 - Nordyne, d.b.a. Westinghouse Heating and Cooling (HVAC)
-and several more

What does Westinghouse produce? Nothing

In 1998, CBS bought the Westinghouse Corporation and immediately sold off all physical assets. From 1998-2021, it licensed the Westinghouse brand to electronic components manufacturers in several industries. Likely made \$100-500mm over 23 years at 100% margin. S&P 500 firms' aggregate (market capitalization - tangible assets)/(market capitalization) grew from 17% in 1976 to 92% in 2026

Case study: Netflix

- 1997: Founded to deliver movies over the internet
- 1999: Pioneered DVD-by-mail
- 2007: Launched streaming
- 2011: July price hike, Sept service split into
 - Netflix: Video streaming
 - “Qwikster”: Rebranded DVD-by-mail service



This case study is ancient but it makes a key point

Qwikster From Netflix: The Worst Product Launch Since New Coke? [OPINION]



September 19, 2011 by Chris Taylor

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Customers fume over Netflix changes



By Doug Gross, CNN

updated 11:49 AM EDT, Tue September 20, 2011 | Filed under: [Web](#)



Jason Gilbert

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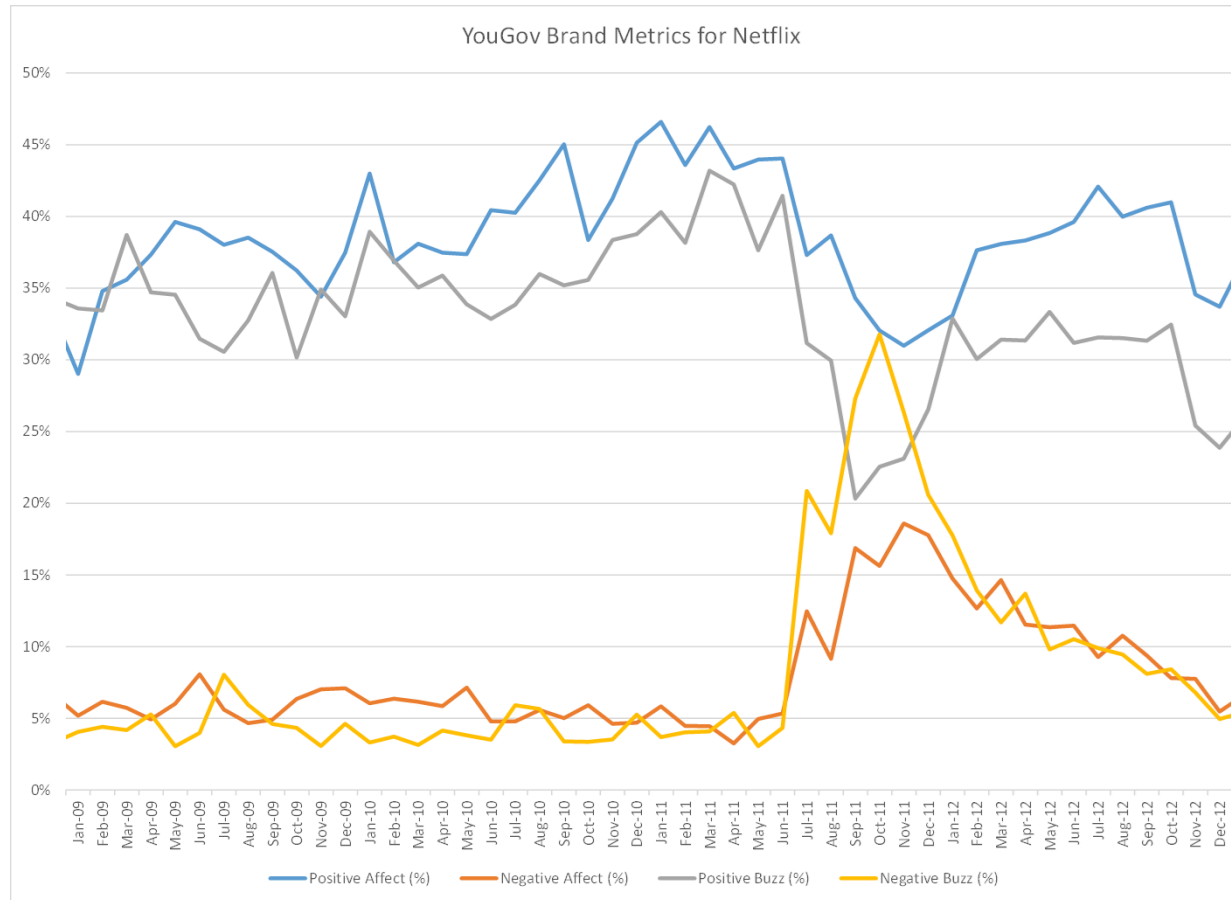


Netflix Announces Qwikster, A DVD Service That Should Die Qwikly

The market reaction was as vicious as I've ever seen



CEO Reed Hastings apologized on video — but said Netflix would proceed with the split anyway, and introduced Qwikster CEO. Guess how many people saw this video on youtube?



These are called monthly brand tracking statistics, summarizing surveys from ~2,400 respondents/month. Affect means, How do you feel about Netflix? Buzz means, What have you heard about Netflix? Answers could be positive, negative, or nothing. First shift was the July price hike; the second was September's rebrand. What do you see?

Netflix, Inc. (NFLX) ★ Watchlist

336.43 -0.25(-0.07%) NASDAQ - As of 4:00PM EST

After Hours: **336.43** 0.00 (0.00%) 04:47pm EST



Note the vertical drop in the stock price on the rebranding date. The second, smaller drop was when they announced they were going ahead with it. Netflix dropped the rebrand shortly after

Netflix Case study: Takeaways

- Brand adjustments require consumer acceptance
 - You own the IP, not the mental real estate
 - Smaller steps encounter less resistance
- Great brands monitor customer perceptions
- If you mis-step, change course – some news is bad news

Coda: DVD-by-mail business fell by 90% from 2010 to 2019, then Netflix announced a gradual phasing out

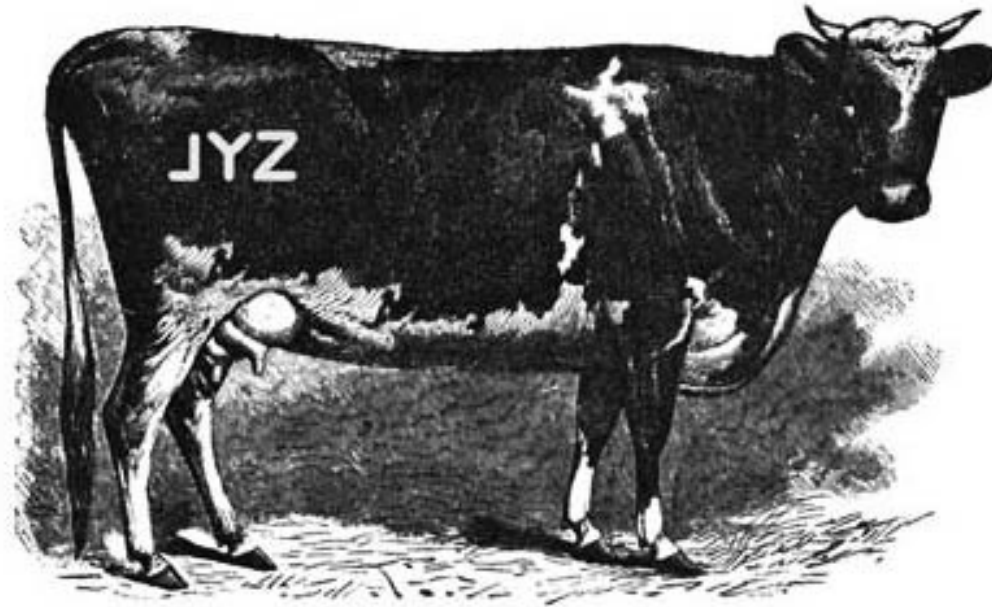


Case study: Kraft Mac n Cheese



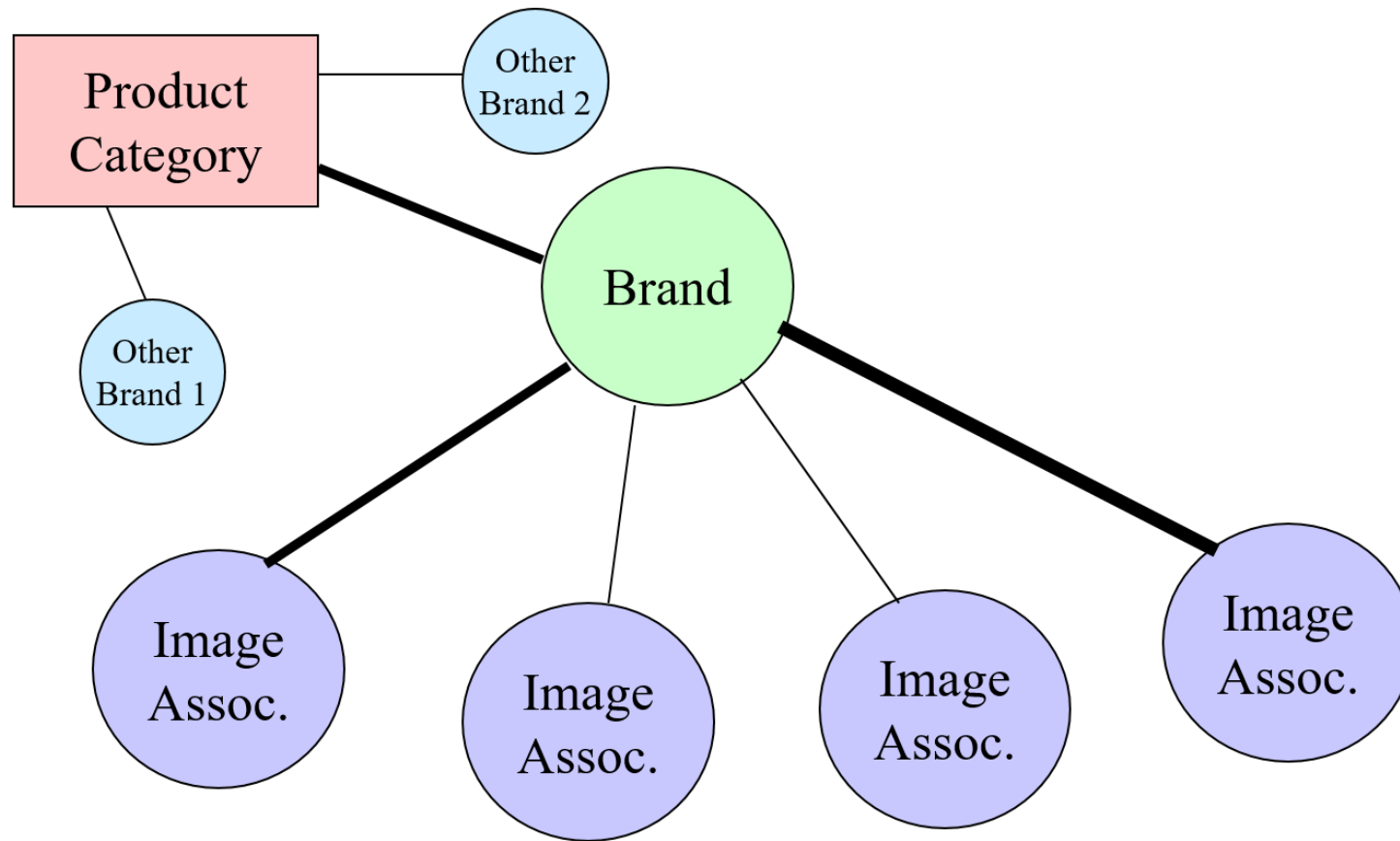
Kraft quietly dropped Yellow 5 & 6, artificial flavors and preservatives—replacing them with paprika, annatto, and turmeric—then announced the change after 50 million boxes had shipped. “We’d invite you to try it, but you already have.”

What is a brand?



CATTLE BRAND.

The word “brand” comes from the Old West; it was a theft deterrent, but it came to mean much more



Memory is an “auto-associative network.” A brand is defined by the network of concepts it connects to in human memory, including the product category, among other associations

What is a brand?

- An *idea* that shifts preference
- Brands reside in consumers' minds
 - Mental real estate: Firm owns trademark but not brand
 - Rooted in reality but reflects consumer experiences, perceptions and idiosyncrasies

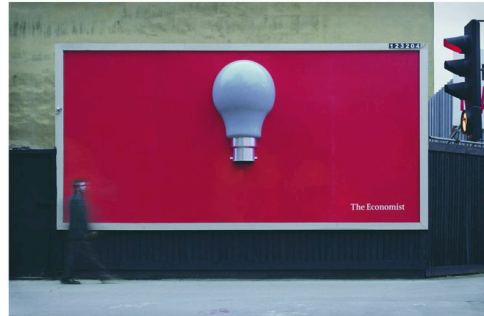
What can be branded? Physical goods, services, retailers, organizations, people, places, activities, even fictional characters

Some auto-associative sequences

Just do...	It (Nike)
I'm lovin'...	It (McDonald's)
Because you're...	Worth it (L'Oreal)
The Ultimate Driving...	Machine (BMW)
A Diamond is...	Forever (De Beers)
Think Outside the	Bun (Taco Bell)
Hulu has live...	sports
What's in your...	Wallet (Capital One)
Red Bull gives you...	Wings
Built Ford...	Tough
Like a Good Neighbor, ...	State Farm is there
America runs on...	Dunkin

Why do firms spend so much to create these associations in consumers' minds? "Search markets" vs. "Selection markets"

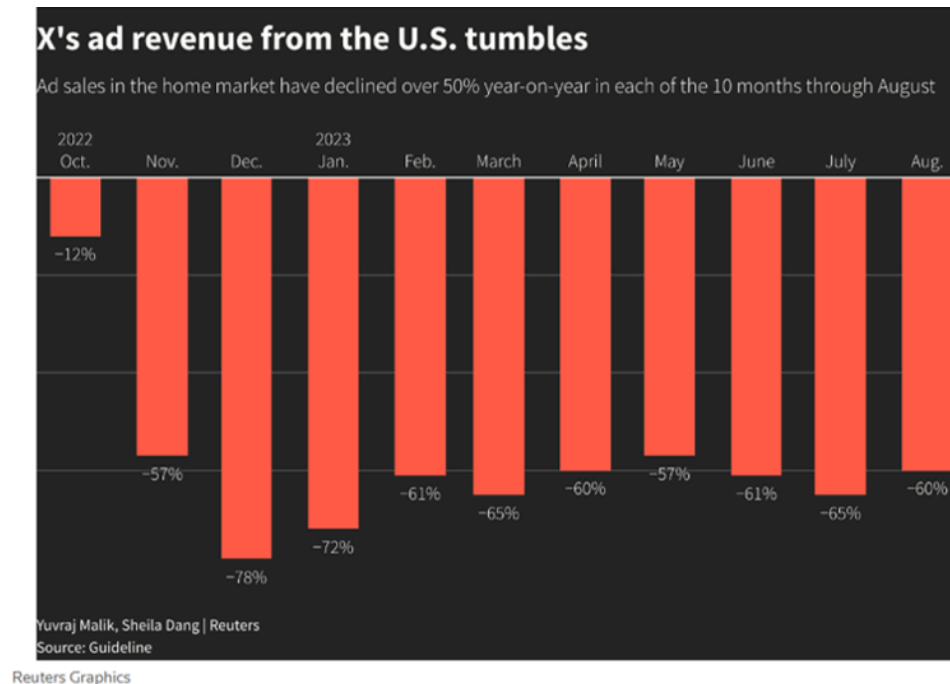
What association?



Firms often seek to influence their brand associations. Here, the *Economist* seeks to influence consumer perceptions of its trademark as a source of bright ideas. When will this work? It depends on what consumers will accept

Brand Safety

- Brands try to avoid unsavory placements – misinformation, hate, violence, obscenity, etc – as they may upset consumers, generate embarrassing publicity, and damage brand associations
- Brands demand advertising-supported platforms maintain “brand safety” via content moderation



Have you ever heard of “brand safety?” When Twitter/X’s new owner announced the shift to a “free speech platform” without content moderation, advertisers asked him to reconsider due to brand safety risks. Earlier ad boycotts included [Facebook](#) and YouTube

How do Brands Work?

1. Credibility
2. Rapid communication
3. Personality & Self-expression
4. Gestalt

1. Credibility



See Size Options

Bayer Aspirin Regimen Low Dose 81mg, Enteric Coated Tablets, 300-Count

\$11.38 (\$0.04/Count) [Subscribe & Save](#)
Save more with monthly Subscribe & Save deliveries.

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Get it by **Friday, Dec 26**

More Buying Choices
\$11.95 new (33 offers)

#1 Best Seller in Aspirin

Show only Bayer Aspirin items

★★★★★ ▾ 273



See Size Options

Kirkland Signature Low Dose Aspirin, 2 bottles - 365-Count Enteric Coated Tablets 81 mg each

\$9.29 (\$0.01/Count) ~~\$13.28~~ [Prime](#)
Get it by **Friday, Dec 26**

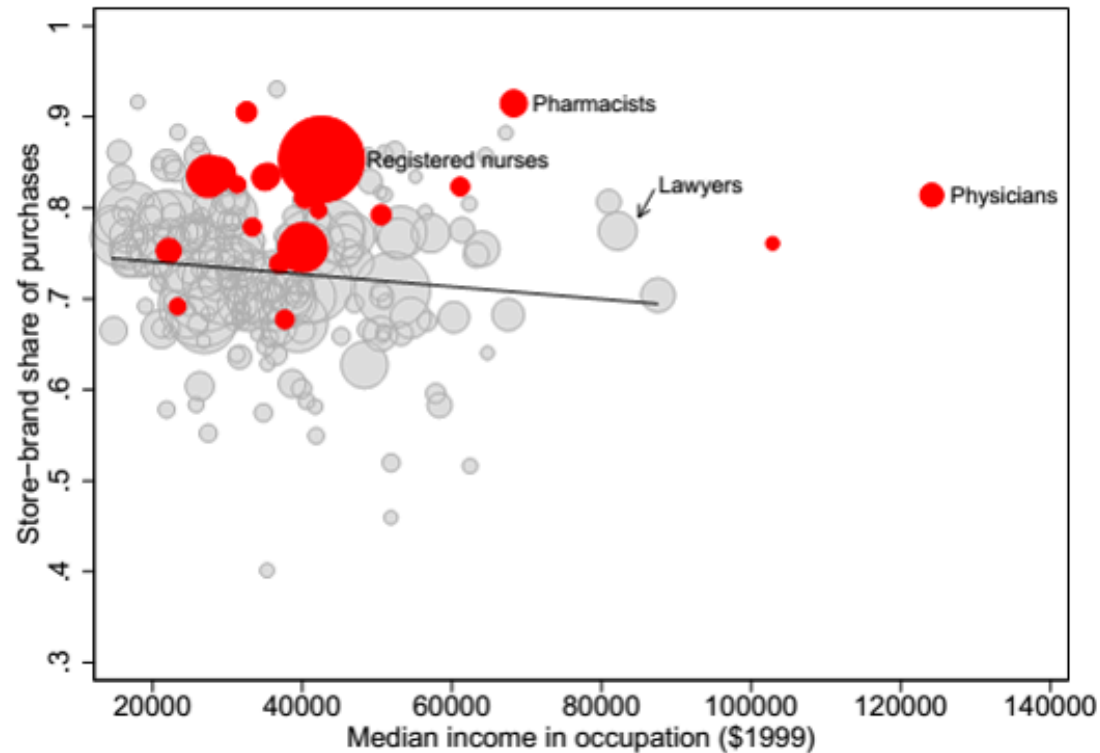
More Buying Choices
\$3.99 new (24 offers)
\$6.22 open box (1 offer)

Show only Kirkland Signature items

★★★★★ ▾ 282

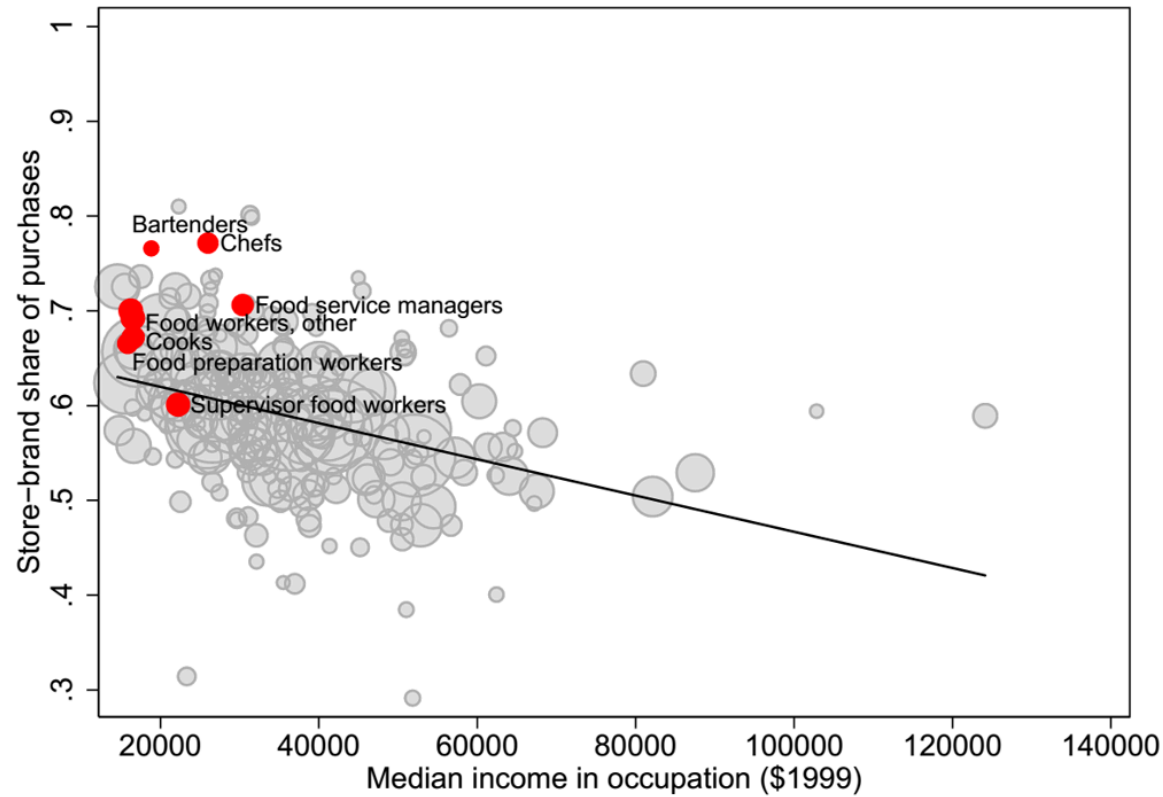
Bayer introduced Aspirin as one of the first patented drugs in the U.S. Aspirin went off patent in the 1950s, and Bayer lost trademark ownership due to genericization after that. Yet Bayer still charges a premium price over nearly identical competitors

Figure 4: Store-brand purchases and occupation, headache remedies



This figure summarizes millions of shopping trips by head-of-household profession. Store-brand shares are high in headache remedies, due to price differentials. Yet household income barely predicts store-brand share. Medical professionals tend to buy store-brands more often; why?

Figure 5: Store-brand purchases and occupation, pantry staples



Similar pattern in commodity foodstuffs (baking soda, plain salt, sugar, etc): Food service professionals more often avoid paying the brand premium due to high product knowledge

Bronnenberg et al. (QJE 2015)

What happens when consumers gain knowledge?

- Bron. et al. (*MkSc* 2020): Intercepted shoppers in-store
- First, a survey, including:
 - Q2: Do you prefer private label or nat'l brand? (NB=0, PL=1)
- Then, a blind brand/PL taste test, followed by
 - Q3: Which did you prefer? (NB=0, PL=1)
 - Q4: Will you buy PL next time?

Some shoppers said they prefer the National Brand, but learned they prefer the Private Label (Q2=0, Q3=1); some said they prefer the Private Label, but learned they prefer the National Brand (Q2=1, Q3=0); the remainder confirmed their expectation. How do you think stated preferences changed in each case?

Bronnenberg, Dube & Sanders (*MkSc* 2020)

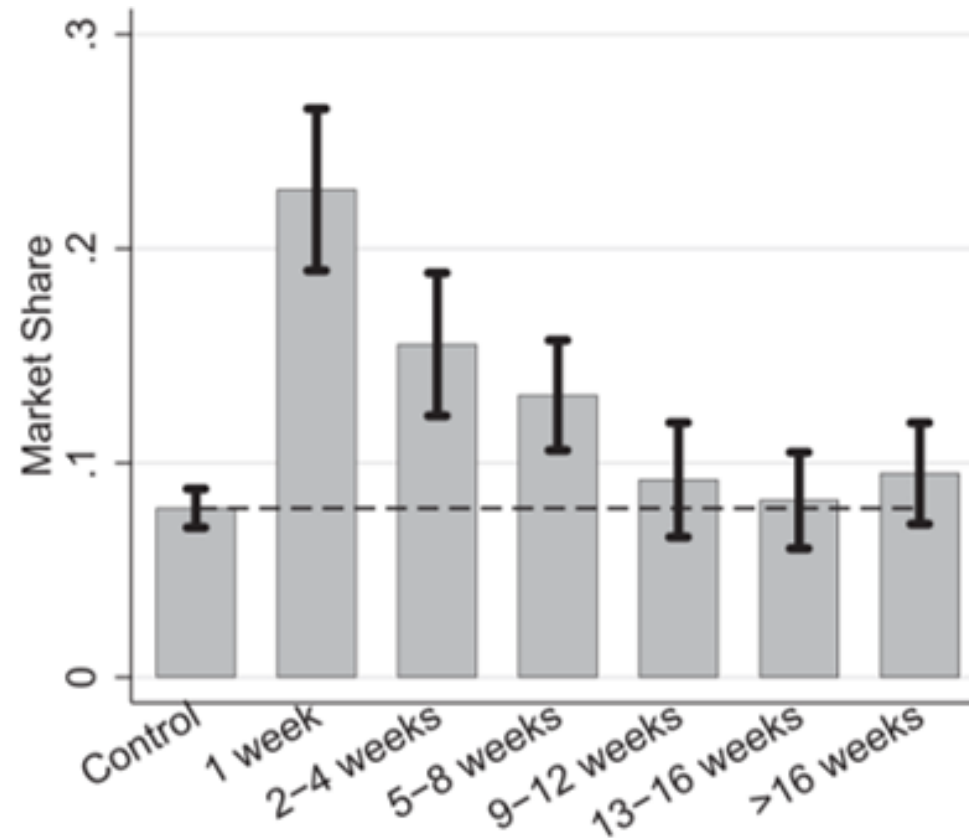
Table 7. Valence of Beliefs Update and Intention to Purchase

		After taste test:		
		Will buy private label next time?		
		No	Yes	Total
Valence of updating during taste test	Negative update	16	36	52
	($Q_2 = 1, Q_3 = 0$)	30.77%	69.23%	100.00%
	Positive neutral	12	225	237
	($Q_2 = 1, Q_3 = 1$)	5.06%	94.94%	100.00%
	Negative neutral	67	69	136
	($Q_2 = 0, Q_3 = 0$)	49.26%	50.74%	100.00%
	Positive update	9	230	239
	($Q_2 = 0, Q_3 = 1$)	3.77%	96.23%	100.00%
	Total	104	560	664
		15.66%	84.34%	100.00%

Notes. The total number of respondents (N) is equal to the number of survey respondents who have a purchase history in the category surveyed. Q_2 – Do you think you will prefer the Private Label brand? (pretest) Q_3 – Did you prefer the Private Label brand?

The intervention had a dramatic effect on stated preferences among those who said they preferred the National Brand, but then learned they preferred the Private Label. Recall that the Private Label is cheaper. Have you ever given yourself a blind taste test to learn your own preferences?

PL Share among Surveyed Shoppers



A cool thing about this study was that survey answers could be linked to frequent shopper cards, so researchers observed subsequent purchases. This is a large impact for a one-time intervention. Why did the effect fade with time?

1. Credibility: Takeaways

- Less-knowledgeable consumers are more likely to pay the brand premium
 - Recognized brands assumed better/safer than alternatives
- Informing consumers can change brand/PL sales
 - Changes may be transitory

Calls back to pricing discussion, how consumers infer unobserved product quality based on price and other signals that would be costly if the firm does not deliver as expected

2. Rapid communication

- Names and graphics
 - facilitate understanding
 - call attention to differentiating attributes
- What is the value proposition?



Target: Busy 8-12-year-olds. Differentiation: no spoon. Intangibles: easy, cool, fun. Helped General Mills overtake Dannon in a mature product category.

3. Personality & Self-expression



Average scores on a scale of 1-10

Who is most & least LIKEABLE?	
	
The GEICO gecko	7.80
Dennis Haysbert's character from Allstate Insurance	7.66
Flo from Progressive Insurance	6.98
Jake from State Farm Insurance	6.78
Professor Burke from Farmers Insurance	6.70
The Aflac duck	6.46
Erin from Esurance	6.06
Mayhem from Allstate Insurance	5.63
The General from General Insurance	5.51
The GEICO caveman	5.12

Geico is in the business of pricing risk. Insurers commonly seek to humanize themselves to seem approachable. As a former Met Life employee told me: “We used to talk about it all the time. We would be nothing without that stupid dog.” (Snoopy)



Does anyone here have a brand tattoo? What do these tattoos say about their wearers? About their brand relationships? We are getting back to the original definition of branding

Brands can

- Express shared identities: UC San Diego, San Diego Padres
- Personify seller attributes, eg Jordan Brand, Kylie Cosmetics
- Adopt a common brand “personality”
 - Sincerity (honest, wholesome): Hallmark, Trader Joe’s, In-n-Out
 - Excitement (daring, youthful): Red Bull, Duolingo
 - Competence (reliable, efficient): Amazon, Google, Toyota, Visa
 - Sophisticated (glamorous, refined): Apple, Rolex, Tiffany
 - Rugged (tough, outdoorsy): Carhartt, Jeep, North Face, Yeti

Best practice: match brand personality to value proposition & target customer’s preferences/aspirations

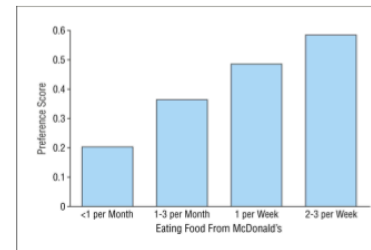
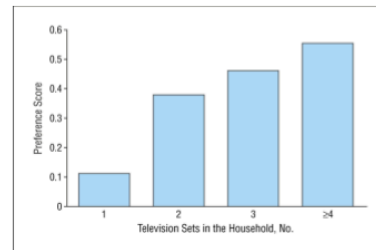
Davies et al. (2018)

4. Gestalt

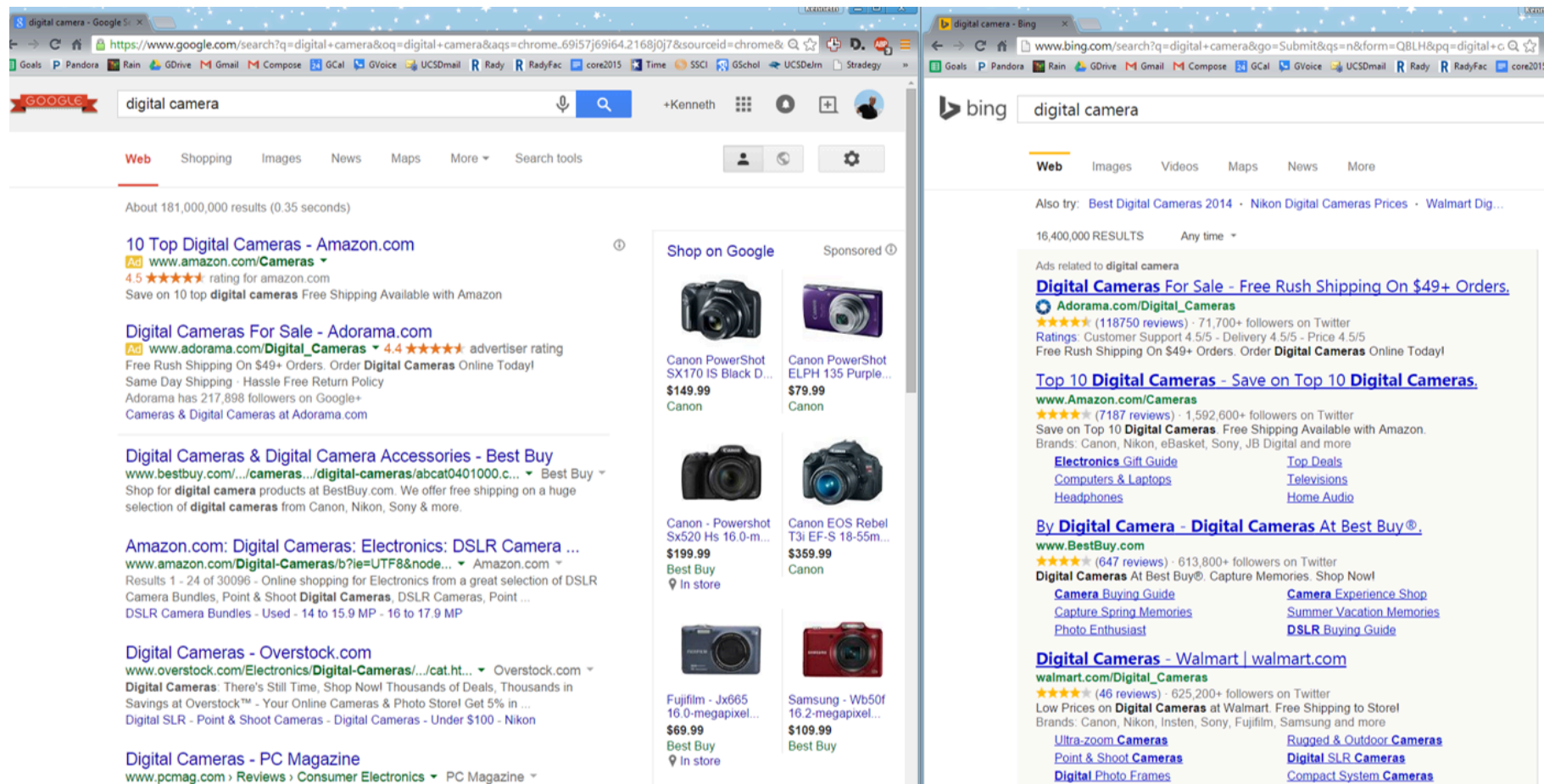
Table 2. Children's Taste Preferences

Food or Drink Item	No. (%)			P Value ^a
	Plain	Taste the Same or No Answer	McDonald's	
Hamburger	22 (36.7)	9 (15.0)	29 (48.3)	.33
Chicken nuggets	11 (18.0)	14 (23.0)	36 (59.0)	<.001
French fries	8 (13.3)	6 (10.0)	46 (76.7)	<.001
Milk or apple juice	13 (21.0)	11 (17.7)	38 (61.3)	<.001
Carrots	14 (23.0)	14 (23.0)	33 (54.1)	.006

^aNonparametricMcNemar test.



We have evidence that branding can complement product consumption. In this study, 3-5 year olds were given identical food in McDonald's branded containers or plain containers, and asked which version tasted better, indicating the McDonald's-branded versions for 4 out of 5 foods. Brand preference increased with TV exposure and brand experience. Neuroscientists gave Coke or Pepsi to subjects in fMRI machines, reporting that "In the brand-cued experiment, brand knowledge for one of the drinks had a dramatic influence on expressed behavioral preferences and on the measured brain responses."



One way Google tracks its own brand strength is to compare preference differentials in its own search results compared to Bing's, vs. Bing's results in Google's format compared to Google's results in Bing's format. Brand dominates content in predicting preference

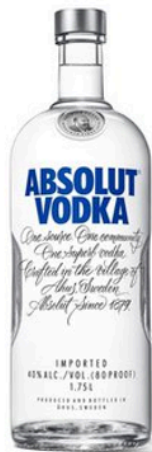
Creating artificial differentiation

Vodka Definition (ATF reg. 27 CFR 5.22):

Neutral spirits so distilled, or so treated after distillation with charcoal or other materials, as to be without distinctive character, aroma, taste or color



\$45.99



\$21.99



The vodka category proves that branding can create differentiation that relieves pricing pressure. It costs around \$5 to make a low-end bottle of vodka, \$15-17 on the high end

Liquid Death's Brand Strategy

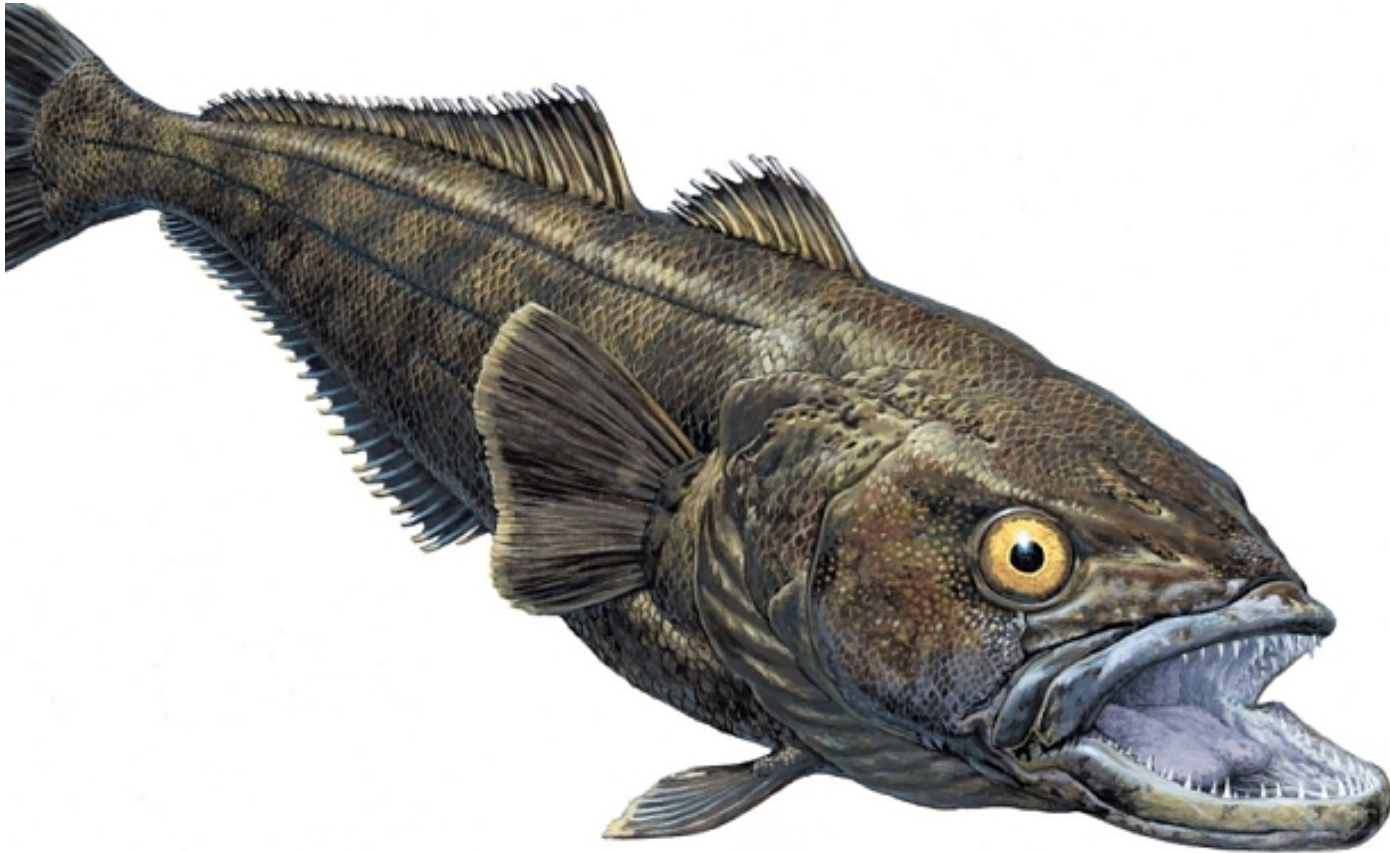
1. Build an entertainment-first brand
 - Comedy is the one thing Coke and Pepsi can't do
2. Prioritize top of funnel (awareness)
3. Be nimble and take risks
4. Problems? Entertain the solution
 - People thought LD sold alcohol, so LD advertised "Drink on the job"
5. Turn your fans into content

Liquid Death founder was a graphic designer at an ad agency inspired by watching festivalgoers reuse an energy drink can to drink water. What's the margin on that water?

Feifer (2025) | a16z (2026)



What's in a name?

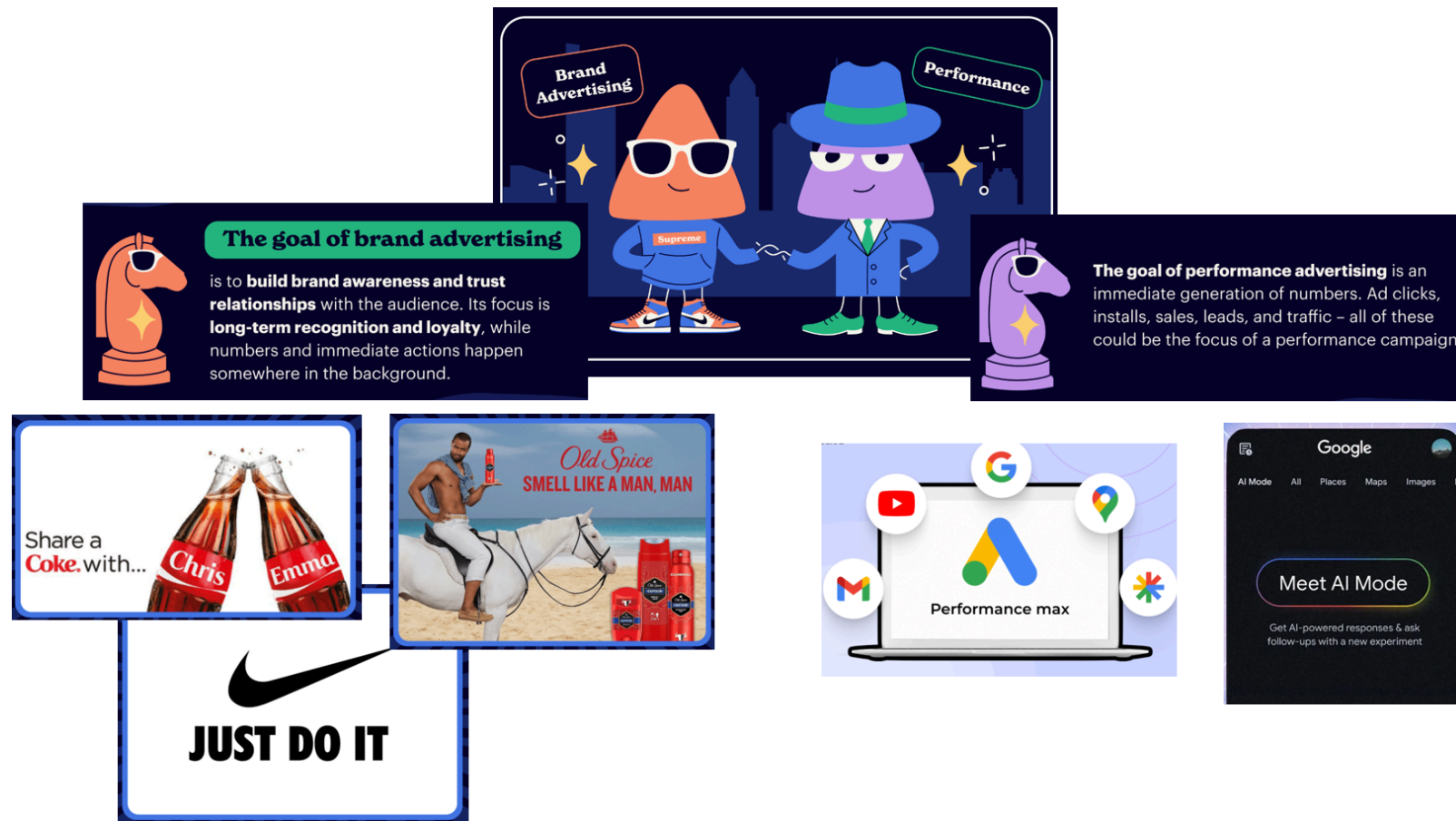


Meet the Patagonian Toothfish. This adorable little monster is a 6' bottom feeder. What would you rename it if you want to start selling it in fancy restaurants?

Advertising

- Brand vs. Performance advertising
- Programmatic advertising
- Does advertising build brands?
- Advertising measurement

Brand Ads vs. Performance Ads?



Brand and performance ads differ in their time horizon, content, measurability, and optimizability. Which approach do you think is better?

Brand is Dead?



Peter Weinberg • Following
Co-Founder, Evidenza | Synthetic Marketing Research
6d • Edited •

"Brand is dead," says famous brand [Scott Galloway](#).

[Jon](#) and I were so appalled by Galloway's recent anti-brand tirade that we dedicated an entire podcast episode to de-bunking it.

Galloway gives great life advice, but his marketing takes are consistently terrible.

First and foremost, Galloway gets his facts wrong.

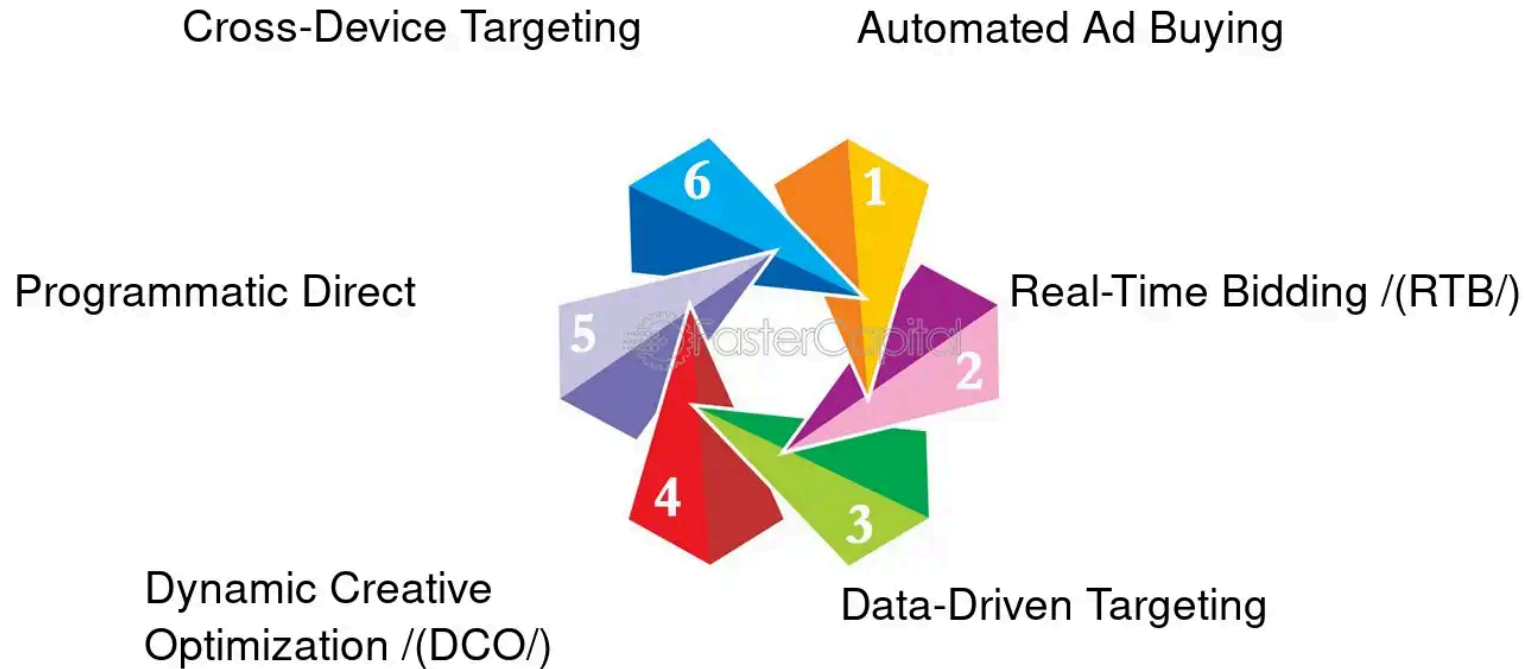
He claims that big companies like Amazon don't invest in advertising, even though Amazon is the largest advertiser on Planet Earth. When [Jon Evans](#) politely pointed that out, Galloway moved the goalposts, claiming that Amazon's spending is all direct response.

That's wrong too --- Amazon spends an estimated \$300MM a year on TV ads. Most big brands invest in brand advertising. God bless [James Hurman](#), who did the math and showed that the "big players" Galloway cites represent 7% of global ad spend.

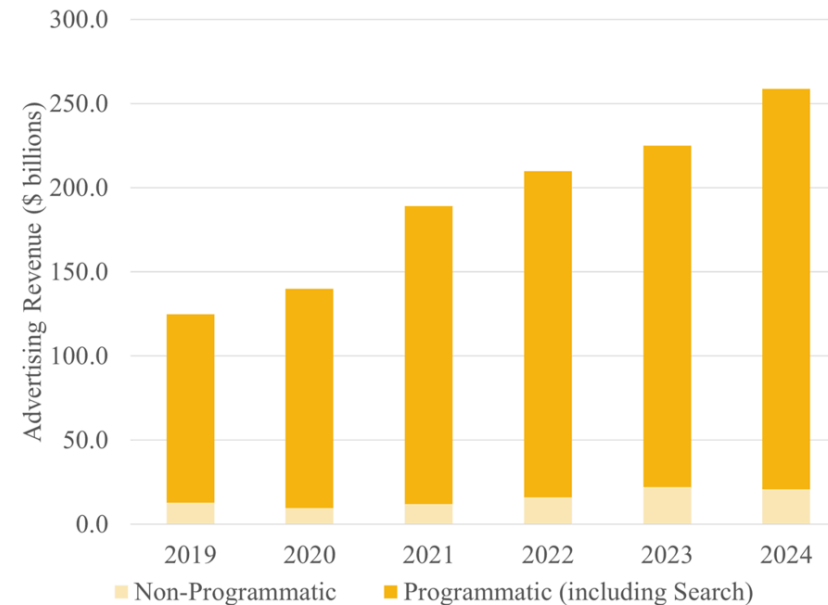
Brands have existed since the dawn of civilization and will exist long after we're all dead. Brands exist because of a quirk in the human brain --- we don't want to waste hours evaluating every product, so we default to fast/easy choices (AKA brands).

We've been hearing takes like this for generations. Every new channel promises to replace brand advertising with something more measurable. So far brand keeps not being dead. Why not?

Introduction to Programmatic Advertising



“Programmatic” means using technology to automate and optimize the ad buying process. Initially, these technologies were developed in the 2000s to allocate unsold ad inventory, but they are now at the top of the food chain



Advertising conversion rates are about 0.1% on average. Programmatic methods promise to leverage data across campaigns, like Waymo (“the world’s most experienced driver”), Netflix recommendations, or Gmail spam filters. More ads placed -> more conversions/non-conversions observed -> more effective placement algorithms

TECH / META

Mark Zuckerberg just declared war on the entire advertising industry



Zuckerberg: But there's still the creative piece, which is basically businesses come to us and they have a sense of what their message is or what their video is or their image, and that's pretty hard to produce and I think we're pretty close.

And the more they produce, the better. Because then, you can test it, see what works. Well, what if you could just produce an infinite number?

Zuckerberg: Yeah, or we just make it for them. I mean, obviously, it'll always be the case that they can come with a suggestion or here's the creative that they want, especially if they really want to dial it in. But in general, we're going to get to a point where you're a business, you come to us, you tell us what your objective is, you connect to your bank account, you don't need any creative, you don't need any targeting demographic, you don't need any measurement, except to be able to read the results that we spit out. I think that's going to be huge, I think it is a redefinition of the category of advertising.

Zuckerberg poses provocative questions. Do platforms still need advertisers to supply advertisements? Would you sign up to connect your bank account?

[The Verge: Zuckerberg on AI and Facebook ads](#)

Should AI make ads?



Quip was accused of using AI to make its ad, so it blogged the production process. How will consumers respond to AI-generated ads?

Modern Retail (2026) | Quip (2026) | Quip ad

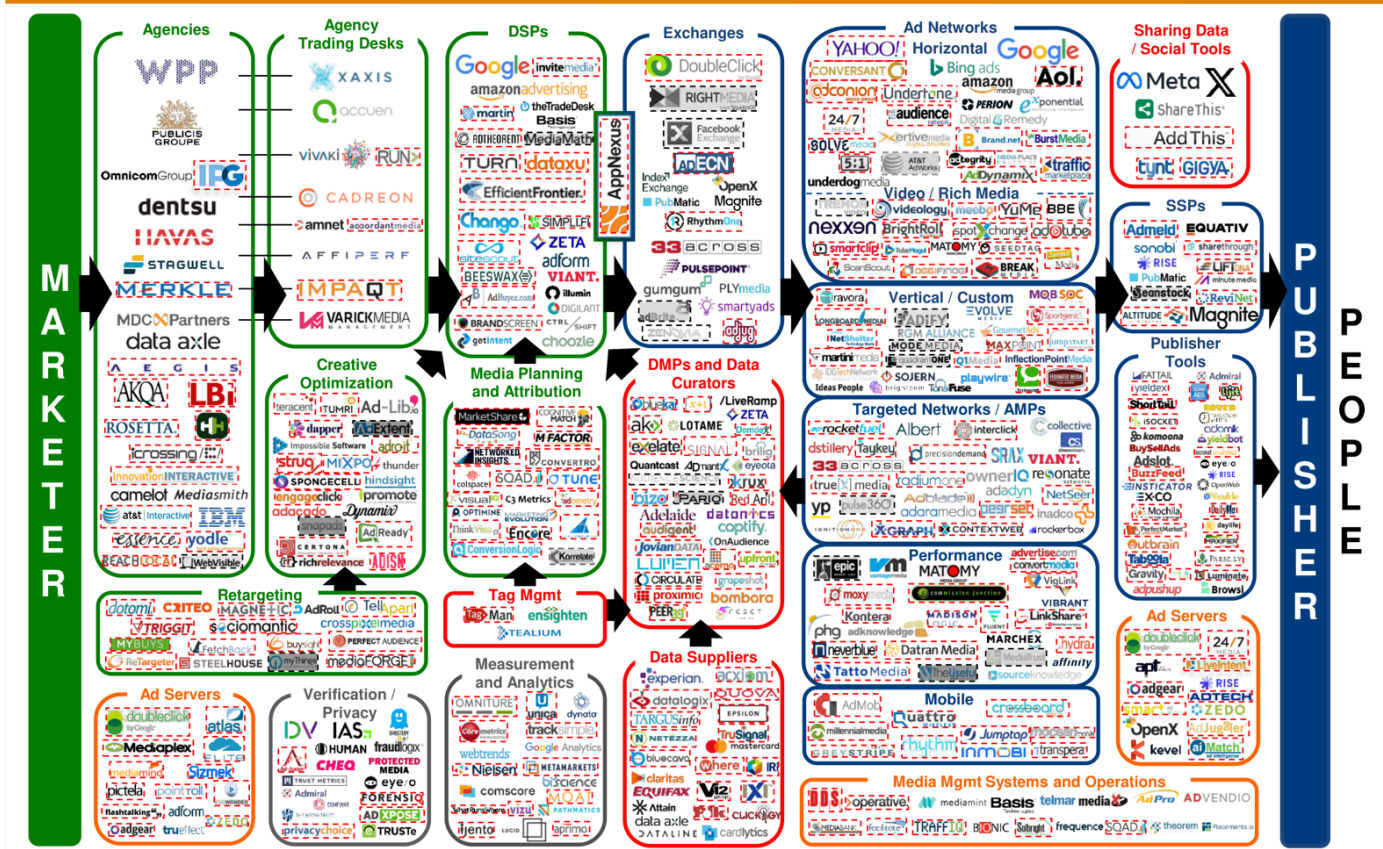
Programmatic downsides

- Infrastructure maximizes conversions, not incremental conversions
 - Attribution/credit “stealing”
- Goliath sellers, unlawful monopolization, enshittification
- Fraud, Algorithmic opacity, Privacy regulations
- Ad blockers: ~25-45% usage, charge ~30% toll
- “Ad Tech Tax”: 20-50% of ad spend

You might ideally want programmatic infrastructure to maximize causal sales lift, but it is not designed that way. Why do you think that is?

[Gordon et al. \(2021, p. 33-37\) overview ad fraud](#)

DISPLAY LUMAscape



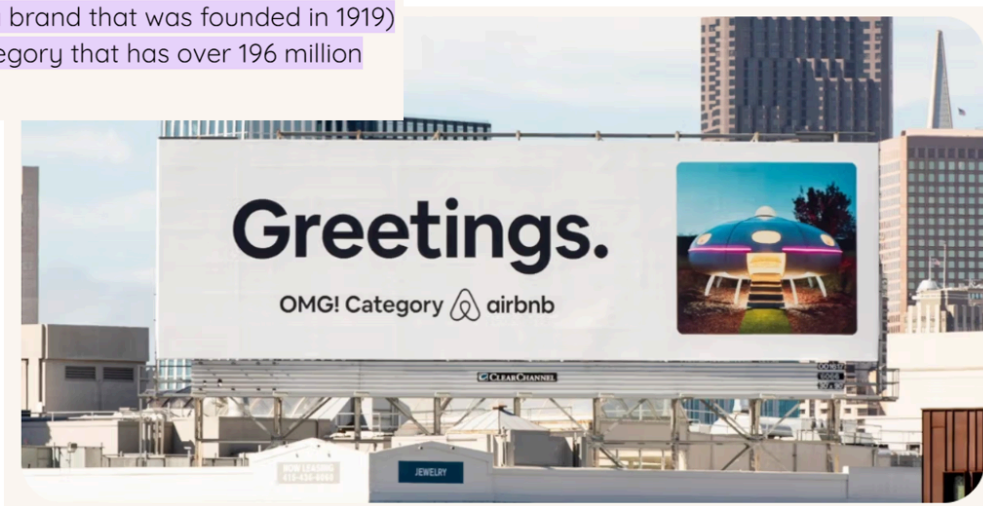
The ad tech tax feeds the ad tech ecosystem. The LUMAscape maps the ad tech supply chain. Each box represents a category of vendors working together to promise the “right ad, right person, right time”

A tectonic shift that paid off, literally 🇧🇷

Two years ago, Airbnb made the shift from a performance-centric marketing strategy, to a brand-centric one.

That meant dialing down spending on search marketing, pay-per-click and other conversion driven activities, while turning up PR, investing in more out-of-home advertising and nurturing previous users to boost retention.

According to Tracksuit data, in the US they have 86% brand awareness (on par with Hilton – a brand that was founded in 1919) in the travel accommodation category that has over 196 million people in it.

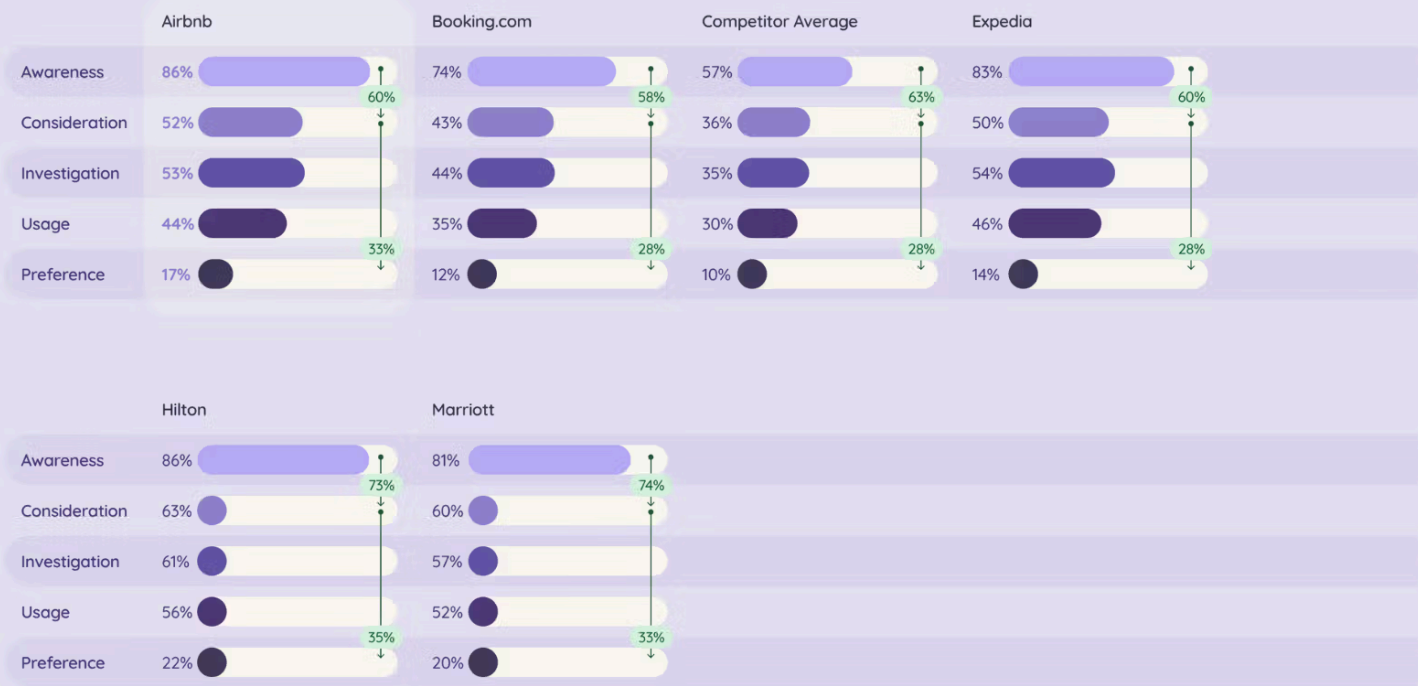


Airbnb shifted from performance ads to brand ads, and said the shift paid off. Should other companies follow suit?

Tracksuit: Airbnb brand investment case | Airbnb brand ad example

Airbnb vs Competitor Brands

Filtered by: Oct 2024 — Dec 2024 | All demographics | Accommodation Services | United States



Airbnb's ad agency argues that Airbnb's high preference:consideration ratio justifies the shift to brand ads... but this viz looks funky

However, Airbnb proves that you need to make brand and performance work together

Though Airbnb is an ideal case study for the power of brand, due to its outspoken advocacy for how brand strategy makes an impact on its profitability, the company also knows one can't exist without the other.

Brian Chesky, Airbnb's CEO, compares performance marketing to a "laser", telling [Lenny's Podcast](#) that performance clearly has its place. "We used to have this metaphor of lasers, flash bulbs and chandeliers. If you want to light up a room, performance marketing is a laser. You don't want to use a bunch of lasers to light up an entire room – you should use a chandelier. That's what brand marketing is.

"Performance marketing, though, doesn't create very good accumulating advantages because it's not an investment."

Airbnb's CFO: We made the right decision

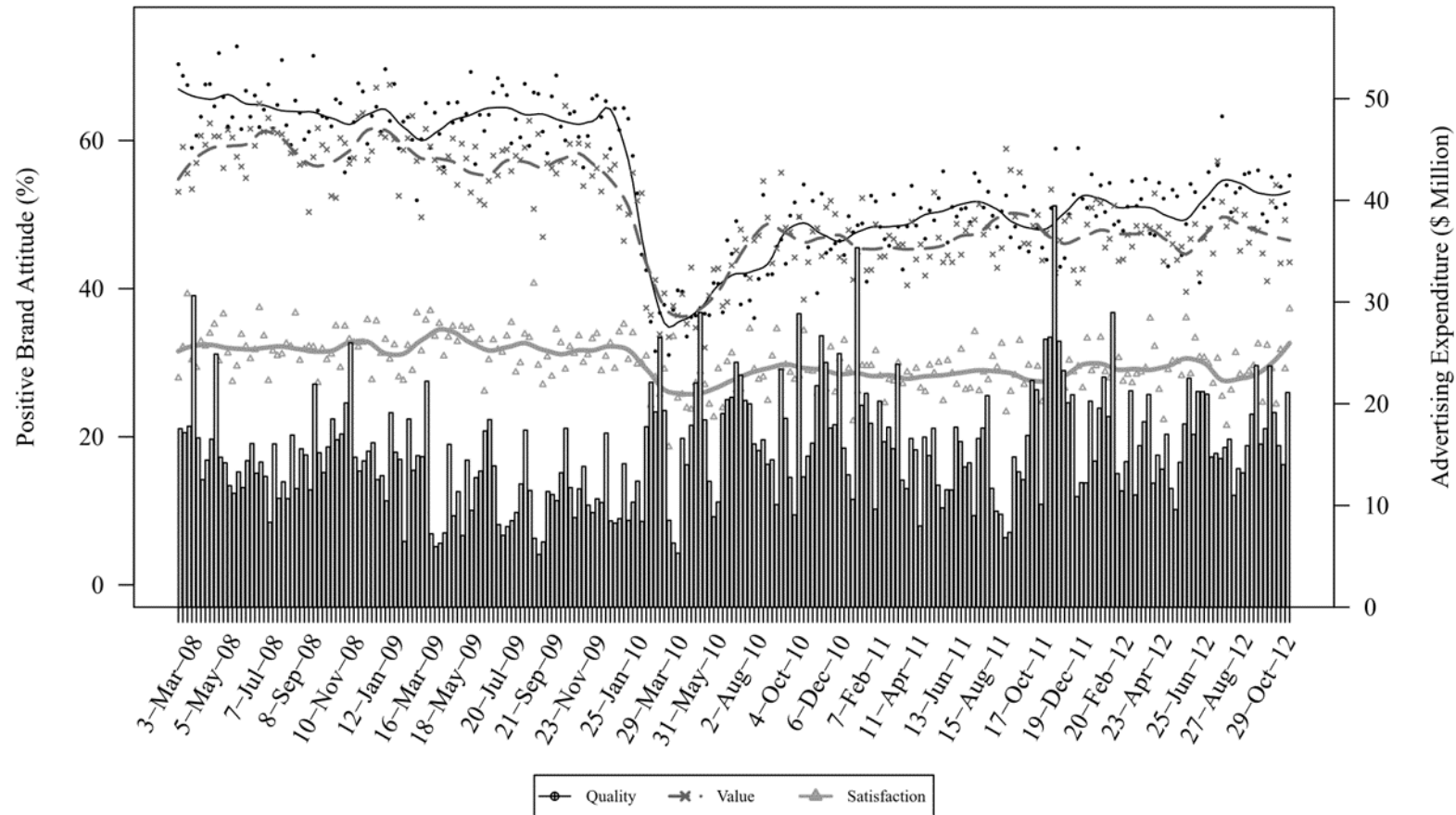
Marketers often tell us about their struggle fighting for more investment (or worse, to avoid budget cuts). Usually, these conversations involve a CFO that's hungry for evidence to be served in support of brand marketing driving commercial gains.

For Airbnb, the financial gains are clear and their CFO, Dave Stephenson, agrees with the company's move to brand-centricity.

Research has proven this time and time again that while performance marketing drives short-term sales activation, it's brand marketing that drives long-term, sustainable growth and profitability.

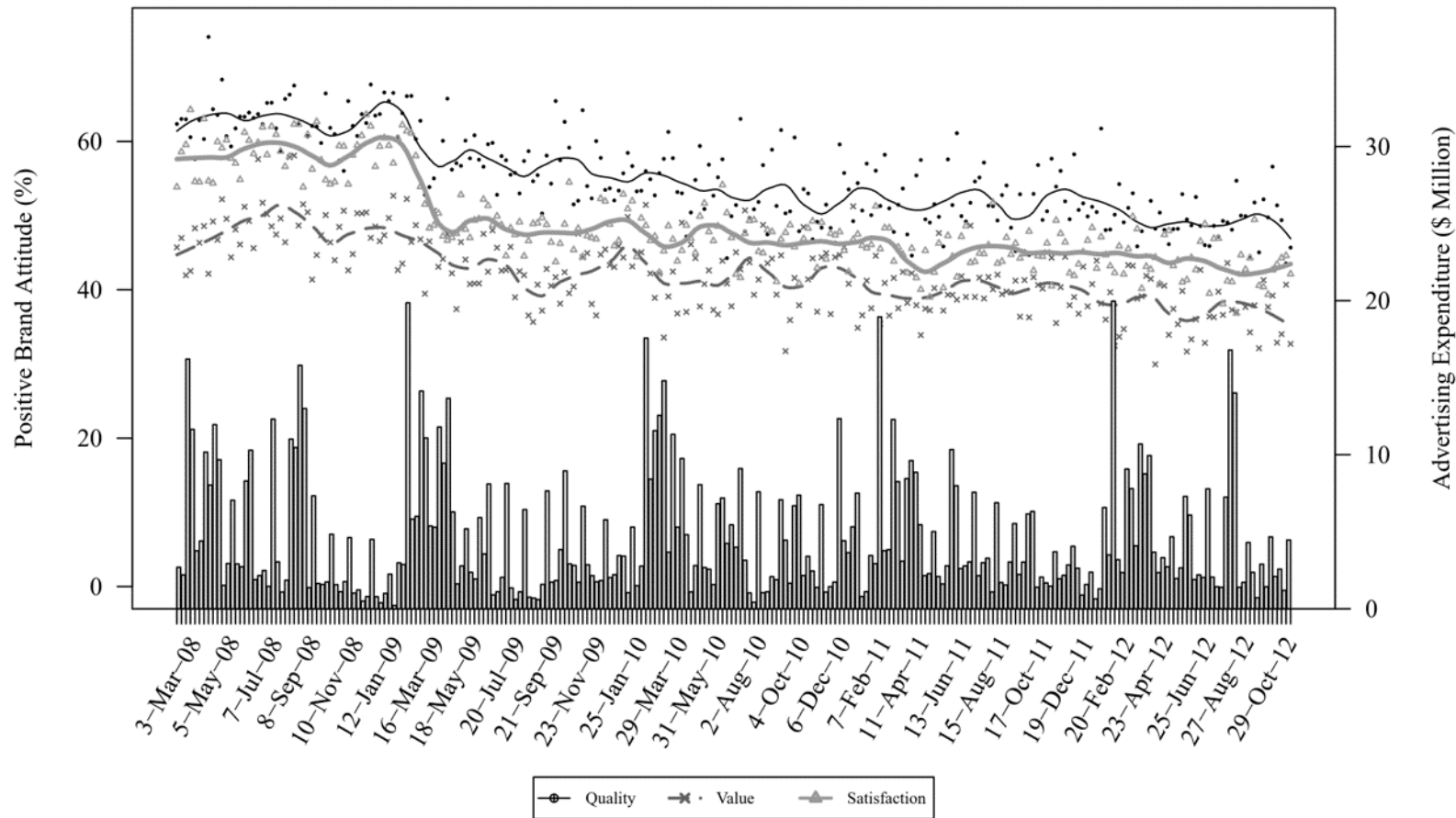
Most large companies run both brand campaigns and performance campaigns, but it is hard to determine the optimal split of advertising budget between them, as we don't have great ways of establishing returns on brand campaigns

Does advertising build brands? Toyota



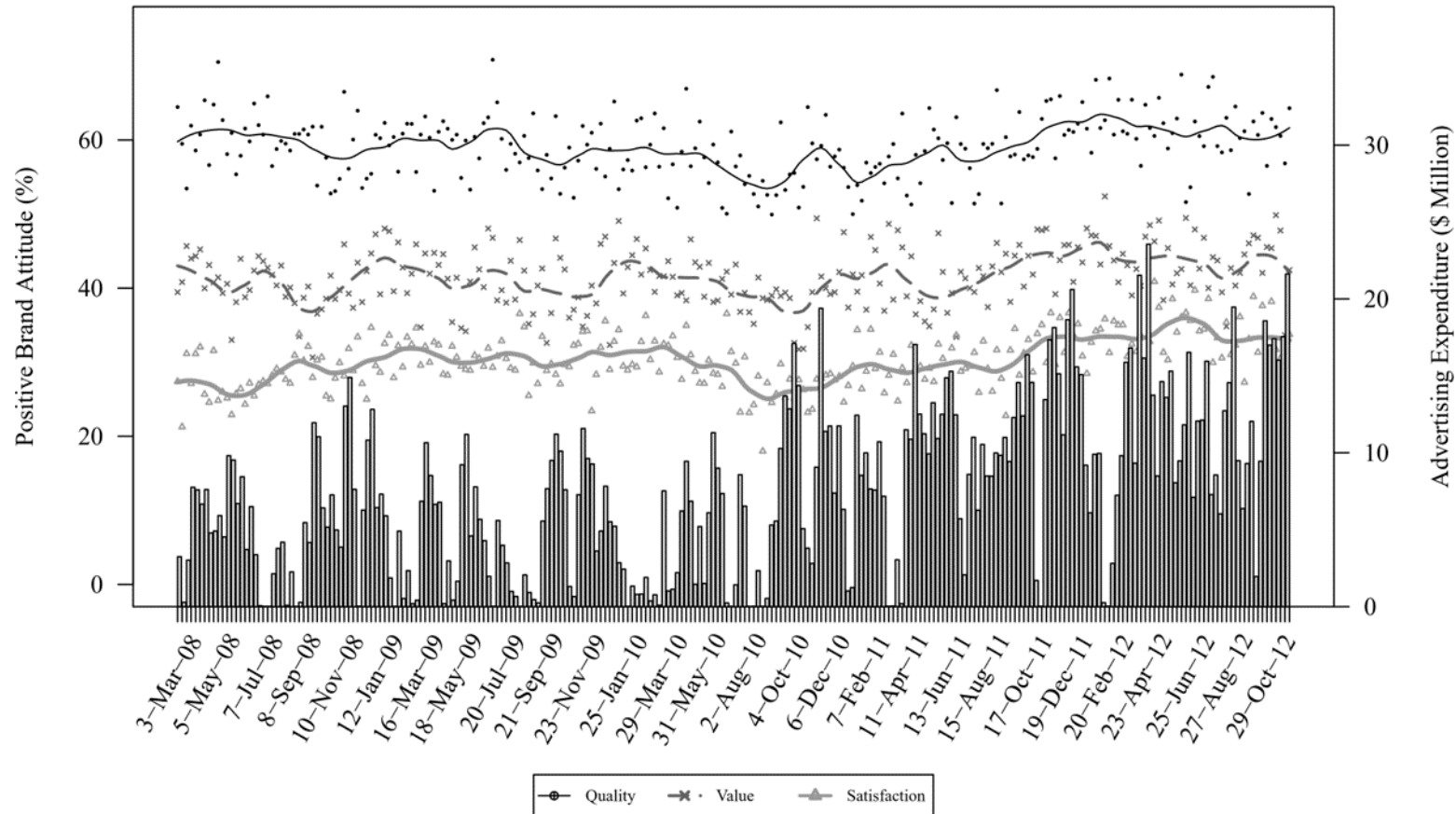
Weekly brand surveys (dots) and moving averages (lines), with weekly ad spend as columns. The 2010 accelerator recall hurt Toyota's brand. Advertising increased afterward and brand attitudes slowly recovered

Does advertising build brands? Coke



Coke's brand metrics slipped as soda lost popularity. Huge Super Bowl ad-spend spikes – not much bounce in brand tracking

Does advertising build brands? Apple



Apple's brand turned upward in late 2010, after Apple introduced the iPad and started advertising more. Ad spending and product quality often correlate

Advertising and brand attitudes

- We regressed brand tracking statistics on brand ad spending of various types, using multiple strategies to control for unobservables
- Findings:
 - Own advertising increases brand perceptions
 - Competitor advertising decreases brand perceptions
 - ⇒ Advertising looks like a prisoner's dilemma
 - Massive data needed to estimate precise effects

We analyzed 10 million brand surveys and \$264B in ad spend by 575 brands in 37 categories over 5 years. This was the largest observational analysis to date, suggesting that advertising is a prisoner's dilemma game that incentivizes brands to over-invest

Du, Joo, Wilbur (2019)

Ads Measurement Approaches

- Marketing Mix Models
- Multi-touch Attribution
- Incrementality Testing
 - Marketers' word for experimental/causal lift
 - A/B/.../n testing, geo-lift tests, going-dark designs
- Quasi-experiments
 - Instrumental variables, synthetic controls

Large marketers usually use multiple methods in parallel. Each approach has known strengths and weaknesses. Ultimately the CFO and CMO decide what gets funded, and how results affect decisions

Tying Things Together

- Brand advertising
 - Should make performance ads more effective
 - Long-term, high-cost, requires strategic commitment
 - Hard to measure incremental returns
 - Possible to measure correlational returns
- Performance advertising
 - Highly targeted, low-cost, short-term, no attachments
 - Definitely possible to measure incremental returns
 - Pray to the algorithmic overlords
- Popular theory: More brand ads in “search markets,” more performance ads in “selection markets”
 - Search markets: Consumer usually searches, eg car insurance
 - Selection markets: Consumer usually chooses from a set, eg grocery store

Brand and performance ads are substitutes in the short run and complements in the long run

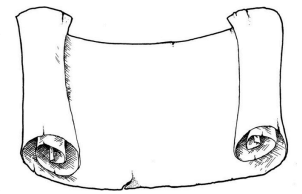
Toolkit: Find wtp for sponsorship

- We can pay a celebrity brand $\$W$ to sponsor our phone. Conjoint estimates indicate the sponsorship will increase phone brand utility from $\hat{\beta}$ to $\hat{\beta}'$. Should we do it?
- Let $\pi_0 = q_j(p_j - c)$ be current contribution
- Let $q'_j(p_j) = N\hat{s}'_j(p_j)$ be celebrity-sponsored demand
- Calculate $\pi_1 = q'_j(p_j)(p_j - c)$
 - $\pi_1 - \pi_0$ is pure demand effect of celebrity association
- Find new optimal price p'_j for demand $q'_j(p_j)$
- Calculate $\pi_2 = q'_j(p'_j)(p'_j - c)$; associate with celebrity if $\pi_2 - \pi_0 > W$
- Discuss: What if the celebrity's threat point is to associate with a competitor?
- Discuss: What if celebrity association persists multiple years?
- Discuss: What if there is a 20% chance that a celebrity scandal reduces $\hat{\beta}'$ to $\hat{\beta}'' < \hat{\beta}$?



Class script

- Two-dimensional grid search to optimize S1 and S2 prices simultaneously
- Assess profit effect of celebrity affiliation with brand, without price reoptimization
- Assess profit effect of celebrity affiliation, with joint price reoptimization

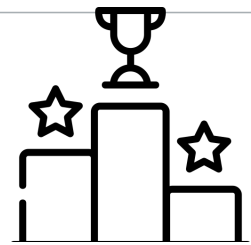


Wrapping up

Competition

- Find the Nash equilibrium in S1-vs-A1 pricing, under two celebrity-endorsement scenarios
- Scenario A: Samsung hires the celebrity (CCE applied to S1 and S2 utilities only)
 - Step 1: For each A1 price in {\$599, \$649, \$699, \$749, \$799, \$849, \$899, \$949, \$999}, find S1's profit-maximizing price using the class dataset and week 6 model `out10`
 - Step 2: For each S1 price in {\$599, \$649, \$699, \$749, \$799, \$849, \$899, \$949, \$999}, find A1's profit-maximizing price (hold all other prices fixed; assume A1 marginal cost = \$450)
 - Step 3: Predict (p_{S1}, p_{A1}) at the intersection of the two best-response curves – this is the Nash equilibrium for Scenario A – and compute profits
- Scenario B: Apple hires the celebrity instead (CCE applied to A1 and A2 utilities)
 - Steps 4-6: Repeat Steps 1-3 for Scenario B
- Plot all four best-response curves in (p_{A1}, p_{S1}) space; mark both Nash equilibria; compare profits across the two scenarios

Do not re-optimize prices of S2 or A2 at any point. This is our most challenging competition of the quarter. We won't ask something this hard on midterm 2.



Recap

- Brands are consumer associations between company assets (products, trademarks, etc) and related concepts
 - Mental real estate: We can influence it but we don't own it
- Brands work by establishing credibility, rapid communication, personality/self-expression and gestalt
- Brand/performance advertising debate trades off LR correlations vs. SR causal effects
- Demand models can quantify returns to brand investments

Going further

- Advertising Measurement by Wilbur (2026)
- Gordon et al (2026): Predicted Incrementality for Experimentation
- Advertising and brand attitudes: Evidence from 575 brands over five years (QME 2019)
- Swaminathan et al. (2020): Branding in a hyperconnected world (JMR)

